Run report: eliminated target-free S_4 against the S_4 target system (lead L1, lemma presentation)

Study `research/target_free_s4_20261018/`, pre-registered in `PREREGISTRATION.md`. **Run** run-20261018T1853Z (2026-10-18 18:53 UTC; queue log 2026-10-18T18:53:28Z to 2026-10-18T20:33:22Z). **Commit** `368d90f34`. **Class:** stage diagnostic, with no `S`, no rho ratio and no speed claim.

What ran

- **Command.** `run.sh`: cells (13,3), (16,4), (17,4), (18,4), (19,4), (20,5), (24,6) in that order, arms gp `[0,0,0,0]`, gp `[0,1,2,3]` and random `[0,0,0,0]`, one process each (below), with `F4_F2_MAX_MATRIX_WORDS = 2^29` (4 GiB step matrices) and 3 rayon threads. The queue stops at the first cell with no finished target-free record.

```
prlimit --as=12000000000 timeout 10800 dreg_eliminated_s4 --cells n:l \
  --basis B --twists T --draws 4 --seed 20261018 --budget-secs 1800
```

- **Arms.** `target_free_s4`: `build_eliminated_s4_system(basis, 1, None, twists)`, 4ℓ unknowns, `n` equations, solved once on gp (deterministic) and on 4 draws on random. `target_s4`: the same with $x_4 = x_R$ (random per draw), 3ℓ unknowns, 4 draws.
- **Measure.** F4 solving degree (field equations included), degree reached, and the top-form deficiency `top_forms − top_rank`, computed before F4.
- **Host.** The 4-core, 15 GB container, which is not isolated (`AGENTS.md` §10). Wall times are practicality notes only.
- **Scorer.** `dreg_score s4_free runs/cell-*.jsonl`, output in `score-output.txt`.

Completion

15 of 18 `runs/cell-*.jsonl` records non-empty; 18 processes exited, 3 with a non-zero status (exit 134 research/target_free_s4_20261018/runs/cell-20-5.gp.tw0000; exit 134 research/target_free_s4_20261018/runs/cell-20-5.gp.tw0123; exit 134 research/target_free_s4_20261018/runs/cell-20-5.random.tw0000). Target-free arm: 30 records, 30 finished, 0 oversize, 0 timed out. Target arm: 60 records, 60 finished, 0 oversize, 0 timed out. Longest single F4 run 406 s; largest recorded step matrix 0.60 GiB. Peak RSS was not recorded.

Queue log (`runs/queue.log`):

```
2026-10-18T18:53:28Z start research/target_free_s4_20261018/runs/cell-13-3.gp.tw0000
2026-10-18T18:53:29Z exit 0 research/target_free_s4_20261018/runs/cell-13-3.gp.tw0000
2026-10-18T18:53:29Z start research/target_free_s4_20261018/runs/cell-13-3.gp.tw0123
2026-10-18T18:53:29Z exit 0 research/target_free_s4_20261018/runs/cell-13-3.gp.tw0123
2026-10-18T18:53:29Z start research/target_free_s4_20261018/runs/cell-13-3.random.tw0000
2026-10-18T18:53:31Z exit 0 research/target_free_s4_20261018/runs/cell-13-3.random.tw0000
2026-10-18T18:53:31Z start research/target_free_s4_20261018/runs/cell-16-4.gp.tw0000
2026-10-18T18:57:09Z exit 0 research/target_free_s4_20261018/runs/cell-16-4.gp.tw0000
2026-10-18T18:57:09Z start research/target_free_s4_20261018/runs/cell-16-4.gp.tw0123
2026-10-18T19:03:57Z exit 0 research/target_free_s4_20261018/runs/cell-16-4.gp.tw0123
2026-10-18T19:03:57Z start research/target_free_s4_20261018/runs/cell-16-4.random.tw0000
2026-10-18T19:19:03Z exit 0 research/target_free_s4_20261018/runs/cell-16-4.random.tw0000
2026-10-18T19:19:03Z start research/target_free_s4_20261018/runs/cell-17-4.gp.tw0000
2026-10-18T19:22:35Z exit 0 research/target_free_s4_20261018/runs/cell-17-4.gp.tw0000
2026-10-18T19:22:35Z start research/target_free_s4_20261018/runs/cell-17-4.gp.tw0123
2026-10-18T19:27:51Z exit 0 research/target_free_s4_20261018/runs/cell-17-4.gp.tw0123
2026-10-18T19:27:51Z start research/target_free_s4_20261018/runs/cell-17-4.random.tw0000
2026-10-18T19:41:58Z exit 0 research/target_free_s4_20261018/runs/cell-17-4.random.tw0000
2026-10-18T19:41:58Z start research/target_free_s4_20261018/runs/cell-18-4.gp.tw0000
2026-10-18T19:46:01Z exit 0 research/target_free_s4_20261018/runs/cell-18-4.gp.tw0000
2026-10-18T19:46:01Z start research/target_free_s4_20261018/runs/cell-18-4.gp.tw0123
2026-10-18T19:52:21Z exit 0 research/target_free_s4_20261018/runs/cell-18-4.gp.tw0123
2026-10-18T19:52:21Z start research/target_free_s4_20261018/runs/cell-18-4.random.tw0000
2026-10-18T20:06:39Z exit 0 research/target_free_s4_20261018/runs/cell-18-4.random.tw0000
2026-10-18T20:06:39Z start research/target_free_s4_20261018/runs/cell-19-4.gp.tw0000
2026-10-18T20:10:14Z exit 0 research/target_free_s4_20261018/runs/cell-19-4.gp.tw0000
2026-10-18T20:10:14Z start research/target_free_s4_20261018/runs/cell-19-4.gp.tw0123
2026-10-18T20:16:34Z exit 0 research/target_free_s4_20261018/runs/cell-19-4.gp.tw0123
2026-10-18T20:16:34Z start research/target_free_s4_20261018/runs/cell-19-4.random.tw0000
2026-10-18T20:29:47Z exit 0 research/target_free_s4_20261018/runs/cell-19-4.random.tw0000
2026-10-18T20:29:47Z start research/target_free_s4_20261018/runs/cell-20-5.gp.tw0000
2026-10-18T20:30:58Z exit 134 research/target_free_s4_20261018/runs/cell-20-5.gp.tw0000
```

```
2026-10-18T20:30:58Z start research/target_free_s4_20261018/runs/cell-20-5.gp.tw0123
2026-10-18T20:32:06Z exit 134 research/target_free_s4_20261018/runs/cell-20-5.gp.tw0123
2026-10-18T20:32:06Z start research/target_free_s4_20261018/runs/cell-20-5.random.tw0000
2026-10-18T20:33:22Z exit 134 research/target_free_s4_20261018/runs/cell-20-5.random.tw0000
2026-10-18T20:33:22Z stop: no finished target-free record at (20:5); larger cells not run
2026-10-18T20:33:22Z queue done
```

Measured values

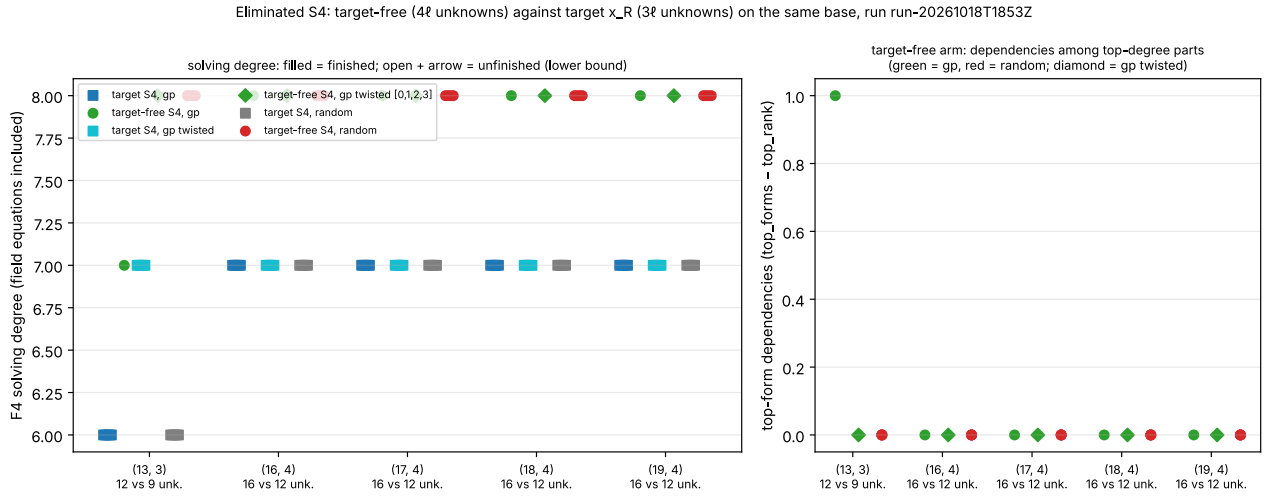


Figure 1: Recorded values from `runs/cell-*.jsonl`, run run-20261018T1853Z. Left: F4 solving degree per draw and arm. Right: top-form dependencies of the target-free arm. Source: `plot.py` (layout only).

Per (cell, basis, twists) readings

(n, ℓ)	basis	twists	target-free solving (≥ = unfinished)	target-free top deficiency	target solving	target top deficiency	target-free value	target median	registered reading	bounded reading
(13, 3)	gp	[0,0,0,0]	7	1	6, 6, 6, 6	4, 4, 4, 4	7	6	higher	not lower
(13, 3)	gp	[0,1,2,3]	8	0	7, 7, 7, 7	0, 0, 0, 0	8	7	higher	not lower
(13, 3)	random	[0,0,0,0]	8, 8, 8, 8	0, 0, 0, 0	6, 6, 6, 6	3, 3, 3, 3	8	6	higher	not lower
(16, 4)	gp	[0,0,0,0]	8	0	7, 7, 7, 7	0, 0, 0, 0	8	7	higher	not lower
(16, 4)	gp	[0,1,2,3]	8	0	7, 7, 7, 7	0, 0, 0, 0	8	7	higher	not lower
(16, 4)	random	[0,0,0,0]	8, 8, 8, 8	0, 0, 0, 0	7, 7, 7, 7	0, 0, 0, 0	8	7	higher	not lower
(17, 4)	gp	[0,0,0,0]	8	0	7, 7, 7, 7	0, 0, 0, 0	8	7	higher	not lower
(17, 4)	gp	[0,1,2,3]	8	0	7, 7, 7, 7	0, 0, 0, 0	8	7	higher	not lower
(17, 4)	random	[0,0,0,0]	8, 8, 8, 8	0, 0, 0, 0	7, 7, 7, 7	0, 0, 0, 0	8	7	higher	not lower
(18, 4)	gp	[0,0,0,0]	8	0	7, 7, 7, 7	1, 1, 1, 1	8	7	higher	not lower
(18, 4)	gp	[0,1,2,3]	8	0	7, 7, 7, 7	0, 0, 0, 0	8	7	higher	not lower
(18, 4)	random	[0,0,0,0]	8, 8, 8, 8	0, 0, 0, 0	7, 7, 7, 7	0, 0, 0, 0	8	7	higher	not lower
(19, 4)	gp	[0,0,0,0]	8	0	7, 7, 7, 7	1, 1, 1, 1	8	7	higher	not lower
(19, 4)	gp	[0,1,2,3]	8	0	7, 7, 7, 7	0, 0, 0, 0	8	7	higher	not lower
(19, 4)	random	[0,0,0,0]	8, 8, 8, 8	0, 0, 0, 0	7, 7, 7, 7	0, 0, 0, 0	8	7	higher	not lower

Registered readings (from `dreg_score s4_free`)

```
Registered verdict (primary): killed. gp (cell, twists) reading lower: []; random reading lower: []; gp readings measured: 10.
Secondary (shape inside target-free): fewer than 2 cells — (13,3)[0000]: gp 7 vs random 8 → shape lower; (16,4)[0000]: gp 8 vs random 8 → equal; (17,4)[0000]: gp 8 vs random 8 → equal; (18,4)[0000]: gp 8 vs random 8 → equal; (19,4)[0000]: gp 8 vs random 8 → equal
P1 (target-free reads higher than target at every measured (cell, basis, twists)): holds — 15 of 15 higher
P2 (gp target-free top deficiency above the random median at every cell with both): fails — 1 of 5 above, 4 equal
P3 (secondary shape effect at ≥ 2 cells): fails
P4 (twisting [0,1,2,3] brings the gp deficiency to the random level or below): holds — (13,3): gp twisted deficiency 0 vs random plain median 0 → at or below random; (16,4): gp twisted deficiency 0 vs random plain median 0 → at or below random; (17,4): gp twisted deficiency 0 vs random plain
```

`median 0 → at or below random; (18,4): gp twisted deficiency 0 vs random plain median 0 → at or below random; (19,4): gp twisted deficiency 0 vs random plain median 0 → at or below random`

Observations (no interpretation)

See `RESULTS.md` for the reading against the registered rules; this report records what ran and what was measured.